

مواد إجباري كليه

=====

COURSE_CODE: 02-24-00101

COURSE_NAME: Linear Algebra

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 1

CREDITS: 3

DESCRIPTION:

Systems of linear equations, Matrices, Determinants, Euclidean spaces, Linear combinations and linear span, Subspaces, Linear independence, Basis and dimension, Rank of a matrix, Inner products, Eigen values and Eigen vectors.

PREREQUISITES:

None

POSTREQUISITES:

02-24-00203 | Numerical Computations

02-24-01301 | Stochastic Processes

02-24-01405 | Social Data Analytics

02-24-01409 | Convex Optimization

02-24-01410 | Non-Linear and Combinatorial Optimization

KEYWORDS:

Linear Algebra, Matrices, Eigenvalues, Vector Spaces, Systems of Equations

=====

=====

COURSE_CODE: 02-24-00102

COURSE_NAME: Calculus

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 1

CREDITS: 3

DESCRIPTION:

Functions and models, Limits and derivatives, Differentiation rules, Applications of differentiation, Integrals and applications of integration, Techniques of integration, Differential equations, Partial differential equations.

PREREQUISITES:

None

POSTREQUISITES:

02-24-01201 | Advanced Calculus

KEYWORDS:

Calculus, Differentiation, Integration, Functions, Differential Equations

=====

=====

COURSE_CODE: 02-24-00103

COURSE_NAME: Introduction to Computer Systems

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 1

CREDITS: 3

DESCRIPTION:

Introduction to computer systems, Representation and manipulation of information, Machine level representation of programs, Introduction to computer organization, Memory hierarchy, System I/O, Introduction to computer networks.

PREREQUISITES:

None

POSTREQUISITES:

02-24-00109 | Introduction to Artificial Intelligence

02-24-00304 | Distributed Processing

02-24-00307 | Operating Systems

02-24-00308 | Computer Networks

KEYWORDS:

Computer Systems, Information Representation, Computer Organization, Memory, I/O

=====

=====

COURSE_CODE: 02-24-00104

COURSE_NAME: Introduction to Data Sciences

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 1

CREDITS: 3

DESCRIPTION:

Introduction to data science, Data science ecosystem, Data representation and manipulation, Tools for data scientists, Data analytics tracks, Case studies using R.

PREREQUISITES:

None

POSTREQUISITES:

02-24-01202 | Data Science Methodology

KEYWORDS:

Data Science, Ecosystem, Data Manipulation, R Language, Analytics

=====

=====

COURSE_CODE: 02-24-00105

COURSE_NAME: Programming I

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 1

CREDITS: 3

DESCRIPTION:

Introduction to programming, Basic programming constructs, Branching and Iteration, Decomposition, Abstractions and Functions, Recursion, Structure types, Mutability, High order functions, Testing and debugging.

PREREQUISITES:

None

POSTREQUISITES:

02-24-00108 | Data Structures and Algorithms

02-24-00110 | Programming II

02-24-00305 | Mobile Programming

02-24-00306 | Web Programming

02-24-00307 | Operating Systems

02-24-01203 | Data Science Tools and Software

02-24-01401 | Big Data Analytics

02-24-01403 | Simulations

KEYWORDS:

Programming, Functions, Recursion, Debugging, Logic

=====

=====

COURSE_CODE: 02-24-00106

COURSE_NAME: Probability and Statistics I

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 2

CREDITS: 3

DESCRIPTION:

Sets, Techniques of counting, Probability spaces, Independence and dependence, Conditional probabilities, Random variables, Expectation, variance, and moments, Moment generating functions, Independence of random variables, Conditional expectation, Discrete and continuous distributions, Joint and marginal distributions.

PREREQUISITES:

None

POSTREQUISITES:

02-24-00201 | Probability and Statistics II

02-24-01403 | Simulations

KEYWORDS:

Probability, Statistics, Random Variables, Distributions, Expectation

=====

=====

COURSE_CODE: 02-24-00107

COURSE_NAME: Discrete Structures

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 2

CREDITS: 3

DESCRIPTION:

Sequences and Summations, Growth of functions, Logic and Predicates, Proof techniques, Recursive relations, Advanced counting techniques, Functions and Relations, Graph and Tree structures, Introduction to number theory: Groups, Rings, and Fields.

PREREQUISITES:

None

POSTREQUISITES:

KEYWORDS:

Discrete Math, Logic, Proofs, Graph Theory, Number Theory

=====

COURSE_CODE: 02-24-00108

COURSE_NAME: Data Structures and Algorithms

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 2

CREDITS: 3

DESCRIPTION:

Arrays, Linear lists, Queues, and Stacks, Tree structures and traversals, Dictionaries and Search trees, Heaps, Tries, Sorting and Searching, Hashing, Basic graph algorithms: Traversals, Minimum Spanning Trees, Shortest Paths.

PREREQUISITES:

02-24-00105 | Programming I

POSTREQUISITES:

02-24-00202 | Introduction to Databases

02-24-00204 | Cloud Computing

02-24-00303 | Algorithm Design

02-24-00304 | Distributed Processing

02-24-01407 | Stream Processing

02-24-01410 | Non-Linear and Combinatorial Optimization

02-24-01413 | Data Compression Techniques

02-24-01414 | Concurrent Algorithms and Data Structures

KEYWORDS:

Data Structures, Algorithms, Sorting, Searching, Graphs

=====

=====

COURSE_CODE: 02-24-00109

COURSE_NAME: Introduction to Artificial Intelligence

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 2

CREDITS: 3

DESCRIPTION:

Reasoning, Intelligent agents, Knowledge representation techniques, Problem solving by searching, Constraint satisfaction problems, Logic programming, Uncertain knowledge and probabilistic reasoning, Planning, Applications.

PREREQUISITES:

02-24-00103 | Introduction to Computer Systems

POSTREQUISITES:

02-24-00205 | Machine Learning

KEYWORDS:

AI, Intelligent Agents, Searching, Logic Programming, Reasoning

=====

=====

COURSE_CODE: 02-24-00110

COURSE_NAME: Programming II

TYPE: إجباري كلية

LEVEL: First Level

SEMESTER: 2

CREDITS: 3

DESCRIPTION:

Object-oriented programming concepts, UML and requirement analysis, Object-oriented design, Encapsulation and information hiding, Separation of interface and implementation, Classes and objects, Methods, Members, Subclasses and inheritance, Polymorphism, Message passing, Operator overloading, Genericity, Programming using threads, Software design patterns.

PREREQUISITES:

02-24-00105 | Programming I

POSTREQUISITES:

02-24-00301 | Software Engineering

KEYWORDS:

OOP, Classes, Inheritance, Polymorphism, Design Patterns

=====

=====

COURSE_CODE: 02-24-00201

COURSE_NAME: Probability and Statistics II

TYPE: إجباري كلية

LEVEL: Second Level

SEMESTER: 3

CREDITS: 3

DESCRIPTION:

Normal distribution, Law of large numbers, Central limit theorem, Distributions derived from Normal distribution: Chi-squared, Student-t, and F distributions, Statistical estimation, Point estimation, Confidence intervals, Test of hypotheses, Fitting straight lines, Analysis of variance, Stochastic models, Poisson processes.

PREREQUISITES:

02-24-00106 | Probability and Statistics I

POSTREQUISITES:

02-24-00206 | Data Mining and Analytics

02-24-01203 | Data Science Tools and Software

02-24-01204 | Regression Analysis

02-24-01301 | Stochastic Processes

02-24-01302 | Design and Analysis of Experiments

02-24-01305 | Survey Methodology

02-24-01306 | Computing Intensive Statistical Methods

02-24-01402 | Introduction to Social Networks

02-24-01405 | Social Data Analytics

02-24-01411 | Multivariate Statistical Analysis

02-24-01412 | Bayesian Statistics

KEYWORDS:

Normal Distribution, Estimation, Hypothesis Testing, ANOVA, Poisson Process

=====

=====

COURSE_CODE: 02-24-00202

COURSE_NAME: Introduction to Databases

TYPE: إجباري كلية

LEVEL: Second Level

SEMESTER: 3

CREDITS: 3

DESCRIPTION:

Information management concepts, Database systems concepts, Data modeling, The relational data model and Relational algebra, Logical database design, Normalization, Query languages, Query optimization, Physical database design, B trees and Indices, Transaction processing concepts, Concurrency control, Recovery.

PREREQUISITES:

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

02-24-01406 | Distributed Data Analysis

02-24-01415 | Distributed Database Systems

02-24-01416 | Advanced Database Systems

KEYWORDS:

Database, SQL, Normalization, Relational Model, Transactions

=====

=====

COURSE_CODE: 02-24-00203

COURSE_NAME: Numerical Computations

TYPE: إجباري كلية

LEVEL: Second Level

SEMESTER: 3

CREDITS: 3

DESCRIPTION:

Matrix manipulation, Simultaneous linear equations and matrix inversion, Vector space and basis, Eigen values and Eigen vectors, Approximation of roots of equations, Error

analysis and Numerical instability, Numerical differentiation and integration, Interpolation and Extrapolation, Least-square approximation, Differential equations.

PREREQUISITES:

02-24-00101 | Linear Algebra

POSTREQUISITES:

02-24-01409 | Convex Optimization

KEYWORDS:

Numerical Analysis, Matrix Manipulation, Root Approximation, Error Analysis, Interpolation

=====

مواد اجباري برنامج

=====

COURSE_CODE: 02-24-01201

COURSE_NAME: Advanced Calculus

TYPE: إجباري برنامج

LEVEL: Second Level

SEMESTER: 3

CREDITS: 3

DESCRIPTION:

Sequences, Series, Power series, Taylor's series, Vector algebra in R^2 and R^3 , Functions of several variables, Partial derivatives, Total differentials, Directional derivatives, Gradients, Lagrange's multiplier, Multiple integrals, Iterated integrals.

PREREQUISITES:

02-24-00102 | Calculus

POSTREQUISITES:

02-24-01407 | Stream Processing

02-24-01409 | Convex Optimization

KEYWORDS:

Advanced Calculus, Taylor Series, Partial Derivatives, Multiple Integrals, Gradients

=====

=====

COURSE_CODE: 02-24-01202

COURSE_NAME: Data Science Methodology

TYPE: إجباري برنامج

LEVEL: Second Level

SEMESTER: 3

CREDITS: 3

DESCRIPTION:

The data science framework, Business understanding, RFM modelling, Data collection and preparation, Structured and unstructured data, Cleaning and Exploration, Data visualization, Market basket analysis, Modeling, Evaluation, Feedback.

PREREQUISITES:

02-24-00104 | Introduction to Data Sciences

POSTREQUISITES:

02-24-01203 | Data Science Tools and Software

02-24-01303 | Data Visualization Tools

KEYWORDS:

Data Science Framework, RFM Modelling, Data Cleaning, Data Exploration

=====

=====

COURSE_CODE: 02-24-00204

COURSE_NAME: Cloud Computing

TYPE: إجباري كلية

LEVEL: Second Level

SEMESTER: 4

CREDITS: 3

DESCRIPTION:

Introduction to Cloud Computing, Cloud computing platforms and infrastructure, Parallel programming in the Cloud, Distributed storage systems, Virtualization, Cloud security, Cloud performance.

PREREQUISITES:

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

02-24-01406 | Distributed Data Analysis

KEYWORDS:

Cloud, Virtualization, Distributed Storage, Cloud Security, Parallel Programming

=====

=====

COURSE_CODE: 02-24-00205

COURSE_NAME: Machine Learning

TYPE: إجباري كلية

LEVEL: Second Level

SEMESTER: 4

CREDITS: 3

DESCRIPTION:

Feature representation, Similarity, Dimensionality reduction, Supervised learning, Regression, Unsupervised learning, Evaluation, Fundamentals of neural networks (Feedforward and Backpropagation), Reinforcement learning, Applications.

PREREQUISITES:

02-24-00109 | Introduction to Artificial Intelligence

POSTREQUISITES:

02-24-01304 | Data Computation and Analysis

02-24-01401 | Big Data Analytics

02-24-01405 | Social Data Analytics

KEYWORDS:

Machine Learning, Neural Networks, Supervised Learning, Regression, Clustering

=====

=====

COURSE_CODE: 02-24-00206

COURSE_NAME: Data Mining and Analytics

TYPE: إجباري كلية

LEVEL: Second Level

SEMESTER: 4

CREDITS: 3

DESCRIPTION:

Introduction to data mining, Data exploration and visualization, Data preprocessing, Classification: concepts, basic techniques and evaluation, advanced methods (SVM, Bayesian networks), Mining frequent patterns and associations, Clustering.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

02-24-01304 | Data Computation and Analysis

02-24-01401 | Big Data Analytics

02-24-01406 | Distributed Data Analysis

02-24-01407 | Stream Processing

KEYWORDS:

Data Mining, Preprocessing, Classification, Clustering, SVM

=====

=====

COURSE_CODE: 02-24-01203

COURSE_NAME: Data Science Tools and Software

TYPE: إجباري برنامج

LEVEL: Second Level

SEMESTER: 4

CREDITS: 3

DESCRIPTION:

Practical data handling and statistical tools, Applications using Python and R, Predictive data analysis software as SAS and Apache Spark, Other data analytics software tools, Use cases in Finance, Media, and Health.

PREREQUISITES:

02-24-00105 | Programming I

02-24-00201 | Probability and Statistics II

02-24-01202 | Data Science Methodology

POSTREQUISITES:

02-24-01303 | Data Visualization Tools

KEYWORDS:

Python, R Language, SAS, Apache Spark, Data Analysis Tools

=====

=====

COURSE_CODE: 02-24-01204

COURSE_NAME: Regression Analysis

TYPE: إجباري برنامج

LEVEL: Second Level

SEMESTER: 4

CREDITS: 3

DESCRIPTION:

Simple linear regression, Multiple regression, Model building and regression diagnostics, One and two factor analysis of variance, Analysis of covariance, Linear model as special case of generalized linear model.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Regression, Linear Model, ANOVA, Diagnostics, Covariance

=====

COURSE_CODE: 02-24-01301

COURSE_NAME: Stochastic Processes

TYPE: إجباري برنامج

LEVEL: Third Level

SEMESTER: 5

CREDITS: 3

DESCRIPTION:

Discrete-time Markov chains, Classification of states, Irreducibility, Periodicity, Convergence theorems and stationary distributions, Poisson process, Point process, Continuous time Markov chain, Birth and death processes, Martingale, Queuing models.

PREREQUISITES:

02-24-00101 | Linear Algebra

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Markov Chains, Poisson Process, Queuing Theory, Martingale, Stochastic Models

=====

COURSE_CODE: 02-24-01302

COURSE_NAME: Design and Analysis of Experiments

TYPE: إجباري برنامج

LEVEL: Third Level

SEMESTER: 5

CREDITS: 3

DESCRIPTION:

Basic experimental designs, Analysis of one-way and two-way layout data, Multiple comparisons, Factorial designs, 2k-factorial designs, Blocking and confounding, Fractional factorial design and nested designs, Representation of results.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Experimental Design, Factorial Design, ANOVA, Blocking, Confounding

=====

COURSE_CODE: 02-24-01303

COURSE_NAME: Data Visualization Tools

TYPE: إجباري برنامج

LEVEL: Third Level

SEMESTER: 5

CREDITS: 3

DESCRIPTION:

Introduction to Information visualization, Visualization tools, Design approaches for visualization, Statistical graphics for univariate, multivariate, temporal and spatial data, Basic design principles, Projections, Rendering techniques, Graphical software packages.

PREREQUISITES:

02-24-01202 | Data Science Methodology

02-24-01203 | Data Science Tools and Software

POSTREQUISITES:

KEYWORDS:

Information Visualization, Statistical Graphics, Rendering, Data Design

=====

COURSE_CODE: 02-24-01304

COURSE_NAME: Data Computation and Analysis

TYPE: إجباري برنامج

LEVEL: Third Level

SEMESTER: 6

CREDITS: 3

DESCRIPTION:

Data storage platforms, Data manipulation, Data cleaning, Data analysis, Dimensionality reduction, Numerical methods for machine learning models, Cluster analysis, Principal component analysis, Kernel methods, Sparse coding, Recommendation systems.

PREREQUISITES:

02-24-00205 | Machine Learning

02-24-00206 | Data Mining and Analytics

POSTREQUISITES:

KEYWORDS:

Data Computation, PCA, Cluster Analysis, Sparse Coding, Recommendation Systems

=====

COURSE_CODE: 02-24-01305

COURSE_NAME: Survey Methodology

TYPE: إجباري برنامج

LEVEL: Third Level

SEMESTER: 6

CREDITS: 3

DESCRIPTION:

Planning of surveys, Questionnaire construction, Methods of data collection, Fieldwork procedures, Sources of errors, Basic ideas of sampling, Simple random sampling, Stratified, systematic, replicated, cluster and quota sampling, Sample size determination.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Surveys, Sampling, Questionnaire, Data Collection, Fieldwork

=====

COURSE_CODE: 02-24-01306

COURSE_NAME: Computing Intensive Statistical Methods

TYPE: إجباري برنامج

LEVEL: Third Level

SEMESTER: 6

CREDITS: 3

DESCRIPTION:

Empirical distribution, Estimation of population distribution, General algorithm of bootstrap method, Bootstrap estimates of standard deviation and bias, Jack-knife method, Bootstrap confidence intervals, Empirical likelihood, EM algorithm.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Bootstrap Method, Jack-knife, EM Algorithm, Statistical Estimation, Empirical Likelihood

=====

COURSE_CODE: 02-24-01401

COURSE_NAME: Big Data Analytics

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 7

CREDITS: 3

DESCRIPTION:

Basics of big data, Big data issues, Big data processing, Streaming big data, Web and social networks data, Classification and Clustering of big data, Modeling big data, Scaling traditional techniques, Use cases.

PREREQUISITES:

02-24-00105 | Programming I

02-24-00205 | Machine Learning

02-24-00206 | Data Mining and Analytics

POSTREQUISITES:

KEYWORDS:

Big Data, Data Streaming, Scaling, Analytics Modeling, Social Network Data

=====

COURSE_CODE: 02-24-01402

COURSE_NAME: Introduction to Social Networks

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 7

CREDITS: 3

DESCRIPTION:

Introduction to online social networks computing, Theoretical foundation, Impact of social networks, Evolution of social networks, Social influence analysis, Link prediction and analysis.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

02-24-01405 | Social Data Analytics

KEYWORDS:

Social Networks, Link Prediction, Influence Analysis, Social Computing

=====

=====

COURSE_CODE: 02-24-01403

COURSE_NAME: Simulations

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 7

CREDITS: 3

DESCRIPTION:

Pseudorandom number generation, Generating discrete and continuous random variables, Simulating discrete events, Statistical analysis of simulated data, Variance reduction, Markov Chain Monte Carlo methods, Stochastic optimization.

PREREQUISITES:

02-24-00105 | Programming I

02-24-00106 | Probability and Statistics I

POSTREQUISITES:

KEYWORDS:

Simulation, MCMC, Random Variables, Discrete Event Simulation, Stochastic Optimization

=====

COURSE_CODE: 02-24-01404

COURSE_NAME: Project I

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 7

CREDITS: 3

DESCRIPTION:

Research skills and technique development through association with a professor in a research effort and/or independent in-depth study in a subject area of common interest.

PREREQUISITES:

None

POSTREQUISITES:

02-24-01408 | Project II

KEYWORDS:

Research Project, Independent Study, Skills Development

=====

=====

COURSE_CODE: 02-24-01405

COURSE_NAME: Social Data Analytics

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 8

CREDITS: 3

DESCRIPTION:

Social data analysis, Influence and centrality in social media, Information diffusion on networks, Topic modeling and sentiment analysis, Identifying social bots, Statistical methods for social data analytics, AI techniques.

PREREQUISITES:

02-24-00101 | Linear Algebra

02-24-00201 | Probability and Statistics II

02-24-00205 | Machine Learning

02-24-01402 | Introduction to Social Networks

POSTREQUISITES:

KEYWORDS:

Sentiment Analysis, Social Media, Information Diffusion, Topic Modeling, Social Bots

=====

COURSE_CODE: 02-24-01406

COURSE_NAME: Distributed Data Analysis

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 8

CREDITS: 3

DESCRIPTION:

Introduction to distributed processing, Parallel computing, Distributed processing platforms, Data analysis using distributed processing platforms, Data mining algorithms in parallel, Case studies using parallel R.

PREREQUISITES:

02-24-00202 | Introduction to Databases

02-24-00204 | Cloud Computing

02-24-00206 | Data Mining and Analytics

POSTREQUISITES:

KEYWORDS:

Distributed Computing, Parallel R, Data Mining, Parallel Processing

=====

COURSE_CODE: 02-24-01407

COURSE_NAME: Stream Processing

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 8

CREDITS: 3

DESCRIPTION:

Dealing with big data online, Data streaming, Stream processing frameworks, Stream processing algorithms, Streaming SQL languages, Analytics while streaming.

PREREQUISITES:

02-24-00101 | Linear Algebra

02-24-00105 | Programming I

02-24-01201 | Advanced Calculus

(Note: Source text has conflicting prerequisites in table vs description; used table listing 02-24-00108, 02-24-00206)

POSTREQUISITES:

KEYWORDS:

Stream Processing, Real-time Analytics, Streaming SQL, Big Data Online

=====

COURSE_CODE: 02-24-01408

COURSE_NAME: Project II

TYPE: إجباري برنامج

LEVEL: Fourth Level

SEMESTER: 8

CREDITS: 3

DESCRIPTION:

The students continue the study performed in the first semester (Project I).

PREREQUISITES:

02-24-01404 | Project I

POSTREQUISITES:

KEYWORDS:

Graduation Project, Research Completion

مواد اختياري كليه

=====

COURSE_CODE: 02-24-00301

COURSE_NAME: Software Engineering

TYPE: اختياري كليه

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Concepts of software development, Software life cycle and process models, Software project management, Requirements engineering, Data and process modeling, Software design techniques, Verification and validation, Evolution, Reliability.

PREREQUISITES:

02-24-00110 | Programming II

POSTREQUISITES:

KEYWORDS:

Software Lifecycle, Process Models, Software Design, Verification, Reliability

=====

COURSE_CODE: 02-24-00302

COURSE_NAME: Systems Analysis and Design

TYPE: اختياري كليه

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Introduction to systems analysis and design, Analysis and design tools, Advanced systems design concepts, Case studies and practical projects, Current trends in systems development.

PREREQUISITES:

None

POSTREQUISITES:

KEYWORDS:

Systems Analysis, Systems Design, Practical Projects, Development Trends

=====

COURSE_CODE: 02-24-00303

COURSE_NAME: Algorithm Design

TYPE: اختياري كلية

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Asymptotic notations, Solving recursive relations, Complexity bounds, Amortized analysis, Randomization, Fundamental design strategies: Divide-and-conquer, Dynamic programming, and Greedy methods, NP-complete problems, Approximation algorithms.

PREREQUISITES:

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

KEYWORDS:

Algorithm Design, Dynamic Programming, Complexity Classes, NP-Complete, Greedy Methods

=====

COURSE_CODE: 02-24-00304

COURSE_NAME: Distributed Processing

TYPE: اختياري كلية

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Parallel and distributed systems architecture models, Distributed communication and message passing (RPC, RMI, MPI), Distributed file systems (GFS, NFS), Synchronization, Fault tolerance, Consistency models, MapReduce, Pig.

PREREQUISITES:

02-24-00103 | Introduction to Computer Systems

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

KEYWORDS:

Distributed Systems, Message Passing, Fault Tolerance, MapReduce, File Systems

=====

COURSE_CODE: 02-24-00305

COURSE_NAME: Mobile Programming

TYPE: اختياري كلية

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Mobile application development frameworks, Design techniques, Methodologies, Android development w/ Java, iOS development w/ Swift, Native React, RESTful apps, Mobile sensors, Security and Privacy.

PREREQUISITES:

02-24-00105 | Programming I

POSTREQUISITES:

KEYWORDS:

Mobile Apps, Android, iOS, React Native, RESTful Services

=====

COURSE_CODE: 02-24-00306

COURSE_NAME: Web Programming

TYPE: اختياري كلية

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Web fundamentals, HTML, CSS, JavaScript basics, Front end programming, PHP, Django, AngularJS, React, responsive web design, Full stack state management, Security pitfalls.

PREREQUISITES:

02-24-00105 | Programming I

POSTREQUISITES:

KEYWORDS:

Web Development, HTML/CSS, JavaScript, Full Stack, React, Django

=====

COURSE_CODE: 02-24-00307

COURSE_NAME: Operating Systems

TYPE: اختياري كلية

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Processes and threads, Synchronization, Scheduling, Memory management and virtual memory, I/O device management, File systems, Virtual machines, Security and protection, Distributed operating systems.

PREREQUISITES:

02-24-00103 | Introduction to Computer Systems

02-24-00105 | Programming I

POSTREQUISITES:

KEYWORDS:

OS, Process Management, Virtual Memory, Scheduling, File Systems

=====

COURSE_CODE: 02-24-00308

COURSE_NAME: Computer Networks

TYPE: اختياري كلية

LEVEL: Third Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Network architecture, layering, and protocols, Application layer, Socket programming, Transport layer (TCP), Principles of reliable data delivery, Network layer: Addressing, Routing and forwarding.

PREREQUISITES:

02-24-00103 | Introduction to Computer Systems

02-24-00605 | - (Required as per text)

POSTREQUISITES:

KEYWORDS:

Networking, TCP/IP, Routing, Protocols, Socket Programming

مواد اختيارية برنامج (Program Electives)

مواد اختياري برنامج

=====

COURSE_CODE: 02-24-01409

COURSE_NAME: Convex Optimization

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Introduction to convex optimization, Basic concepts for convex functions and sub-gradients, Gradient and sub-gradient methods, Accelerated proximal gradient methods, Lagrangian duals, Splitting algorithms.

PREREQUISITES:

02-24-00101 | Linear Algebra

02-24-00203 | Numerical Computations

02-24-01201 | Advanced Calculus

POSTREQUISITES:

KEYWORDS:

Convex Optimization, Lagrangian Duals, Gradient Methods, Proximal Gradient

=====

COURSE_CODE: 02-24-01410

COURSE_NAME: Non-Linear and Combinatorial Optimization

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Optimization techniques for constrained unidimensional and multidimensional problems, Analysis techniques, Matching, Unimodular matrices, Matroids, Max-flows and min-cuts, Heuristic techniques for hard problems.

PREREQUISITES:

02-24-00101 | Linear Algebra

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

KEYWORDS:

Non-Linear Optimization, Combinatorial Optimization, Matroids, Max-Flow, Heuristics

=====

COURSE_CODE: 02-24-01411

COURSE_NAME: Multivariate Statistical Analysis

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Distribution theory, Multivariate normal distribution, Hotelling's T2 and Wishart distributions, Inference on mean and covariance, Principal components and canonical correlation, Factor analysis, Discrimination.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Multivariate Analysis, Principal Components, Factor Analysis, Wishart Distribution

=====

COURSE_CODE: 02-24-01412

COURSE_NAME: Bayesian Statistics

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Bayes' theorem, estimation, hypothesis testing, prior distributions, likelihood, predictive distributions. Bayesian computation: numerical approximation, posterior simulation, MCMC, hierarchical linear models.

PREREQUISITES:

02-24-00201 | Probability and Statistics II

POSTREQUISITES:

KEYWORDS:

Bayesian Inference, MCMC, Prior/Posterior Distribution, Hierarchical Models

=====

COURSE_CODE: 02-24-01413

COURSE_NAME: Data Compression Techniques

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Basics of information theory, Lossless and lossy compression, Huffman coding, LZW coding, Entropy coding, Text compression, Video and audio compression, DNA sequence compression.

PREREQUISITES:

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

KEYWORDS:

Data Compression, Huffman Coding, LZW, Lossless/Lossy, Information Theory

=====

COURSE_CODE: 02-24-01414

COURSE_NAME: Concurrent Algorithms and Data Structures

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Mutual exclusion, Correctness conditions for concurrent datatypes, Foundations of shared memory, Synchronization methods, Implementing locks and monitors, concurrent linked lists, queues, and hash tables.

PREREQUISITES:

02-24-00108 | Data Structures and Algorithms

POSTREQUISITES:

KEYWORDS:

Concurrency, Mutual Exclusion, Synchronization, Locks, Shared Memory

=====

COURSE_CODE: 02-24-01415

COURSE_NAME: Distributed Database Systems

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Distributed database architectures, Distributed data storage and indexing, Distributed and parallel query processing/optimization, Distributed transaction management, Concurrency control.

PREREQUISITES:

02-24-00202 | Introduction to Databases

POSTREQUISITES:

KEYWORDS:

Distributed DB, Query Optimization, Transaction Management, Parallel Databases

=====

COURSE_CODE: 02-24-01416

COURSE_NAME: Advanced Database Systems

TYPE: اختياري برنامج

LEVEL: Fourth Level

SEMESTER: -

CREDITS: 3

DESCRIPTION:

Transaction handling, Advanced SQL, Triggers and stored procedures, Client/server applications, Efficient storing and recovering, Hybrid and object-oriented databases.

PREREQUISITES:

02-24-00202 | Introduction to Databases

POSTREQUISITES:

KEYWORDS:

Advanced SQL, Triggers, Stored Procedures, Object-Oriented DB, Client-Server

=====

SECTION_TYPE: STUDY_PLAN

PROGRAM: Data Science

FIRST LEVEL - SEMESTER 1

REQUIRED COURSES:

02-24-00101 | Linear Algebra

02-24-00102 | Calculus

02-24-00103 | Introduction to Computer Systems

02-24-00104 | Introduction to Data Sciences

02-24-00105 | Programming I

0200000XX | Critical Thinking

SUMMARY:

Required Courses Credits = 17

Program Elective Credits = 0

Faculty Elective Credits = 0

University Elective Credits = 0

=====

FIRST LEVEL - SEMESTER 2

REQUIRED COURSES:

02-24-00106 | Probability and Statistics I

02-24-00107 | Discrete Structures

02-24-00108 | Data Structures and Algorithms

02-24-00109 | Introduction to Artificial Intelligence

02-24-00110 | Programming II

0200000XX | Innovation & Entrepreneurship

SUMMARY:

Required Courses Credits = 17

Program Elective Credits = 0

Faculty Elective Credits = 0

University Elective Credits = 0

=====

SECOND LEVEL - SEMESTER 3

REQUIRED COURSES:

02-24-00201 | Probability and Statistics II

02-24-00202 | Introduction to Databases

02-24-00203 | Numerical Computations

02-24-01201 | Advanced Calculus

02-24-01202 | Data Science Methodology

SUMMARY:

Required Courses Credits = 15

Program Elective Credits = 0

Faculty Elective Credits = 0

University Elective Credits = 2

=====

SECOND LEVEL - SEMESTER 4

REQUIRED COURSES:

02-24-00204 | Cloud Computing

02-24-00205 | Machine Learning

02-24-00206 | Data Mining and Analytics

02-24-01203 | Data Science Tools and Software

02-24-01204 | Regression Analysis

SUMMARY:

Required Courses Credits = 15

Program Elective Credits = 0

Faculty Elective Credits = 0

University Elective Credits = 2

=====

THIRD LEVEL - SEMESTER 5

REQUIRED COURSES:

02-24-01301 | Stochastic Processes

02-24-01302 | Design and Analysis of Experiments

02-24-01303 | Data Visualization Tools

SUMMARY:

Required Courses Credits = 9

Program Elective Credits = 0

Faculty Elective Credits = 6

University Elective Credits = 2

=====

THIRD LEVEL - SEMESTER 6

REQUIRED COURSES:

02-24-01304 | Data Computation and Analysis

02-24-01305 | Survey Methodology

02-24-01306 | Computing Intensive Statistical Methods

SUMMARY:

Required Courses Credits = 9

Program Elective Credits = 0

Faculty Elective Credits = 6

University Elective Credits = 0

=====

FOURTH LEVEL - SEMESTER 7

REQUIRED COURSES:

02-24-01401 | Big Data Analytics

02-24-01402 | Introduction to Social Networks

02-24-01403 | Simulations

02-24-01404 | Project I

SUMMARY:

Required Courses Credits = 12

Program Elective Credits = 6

Faculty Elective Credits = 0

University Elective Credits = 2

=====

FOURTH LEVEL - SEMESTER 8

REQUIRED COURSES:

02-24-01405 | Social Data Analytics

02-24-01406 | Distributed Data Analysis

02-24-01407 | Stream Processing

02-24-01408 | Project II

SUMMARY:

Required Courses Credits = 12

Program Elective Credits = 6

Faculty Elective Credits = 0

University Elective Credits = 2

STUDY PLAN SUMMARY

COLLEGE REQUIRED COURSES (48 Credits)

First Level - Semester 1 (17 Credits)

- Linear Algebra | 02-24-00101
- Calculus | 02-24-00102
- Introduction to Computer Systems | 02-24-00103
- Introduction to Data Sciences | 02-24-00104
- Programming I | 02-24-00105
- Critical Thinking | 0200000XX

Required Courses Count: 6

University Elective Credits: 0

Faculty Elective Credits: 0

Program Elective Credits: 0

=====

First Level - Semester 2 (17 Credits)

- Probability and Statistics I | 02-24-00106
- Discrete Structures | 02-24-00107
- Data Structures and Algorithms | 02-24-00108
- Introduction to Artificial Intelligence | 02-24-00109
- Programming II | 02-24-00110
- Innovation & Entrepreneurship | 0200000XX

Required Courses Count: 6

University Elective Credits: 0

Faculty Elective Credits: 0

Program Elective Credits: 0

=====

Second Level - Semester 3 (17 Credits)

- Probability and Statistics II | 02-24-00201
- Introduction to Databases | 02-24-00202
- Numerical Computations | 02-24-00203
- Advanced Calculus | 02-24-01201
- Data Science Methodology | 02-24-01202

Required Courses Count: 5

University Elective Credits: 2

Faculty Elective Credits: 0

Program Elective Credits: 0

=====

Second Level - Semester 4 (17 Credits)

- Cloud Computing | 02-24-00204
- Machine Learning | 02-24-00205
- Data Mining and Analytics | 02-24-00206
- Data Science Tools and Software | 02-24-01203
- Regression Analysis | 02-24-01204

Required Courses Count: 5

University Elective Credits: 2

Faculty Elective Credits: 0

Program Elective Credits: 0

=====

Third Level - Semester 5 (17 Credits)

- Stochastic Processes | 02-24-01301
- Design and Analysis of Experiments | 02-24-01302
- Data Visualization Tools | 02-24-01303

Required Courses Count: 3
Faculty Elective Credits: 6
University Elective Credits: 2
Program Elective Credits: 0

=====

Third Level - Semester 6 (15 Credits)

- Data Computation and Analysis | 02-24-01304
- Survey Methodology | 02-24-01305
- Computing Intensive Statistical Methods | 02-24-01306

Required Courses Count: 3
Faculty Elective Credits: 6
University Elective Credits: 0
Program Elective Credits: 0

=====

Fourth Level - Semester 7 (18 Credits)

- Big Data Analytics | 02-24-01401
- Introduction to Social Networks | 02-24-01402
- Simulations | 02-24-01403
- Project I | 02-24-01404

Required Courses Count: 4
Program Elective Credits: 6
University Elective Credits: 2

=====

Fourth Level - Semester 8 (18 Credits)

- Social Data Analytics | 02-24-01405

- Distributed Data Analysis | 02-24-01406
- Stream Processing | 02-24-01407
- Project II | 02-24-01408

Required Courses Count: 4

Program Elective Credits: 6

University Elective Credits: 2

=====

course_summary

FACULTY ELECTIVES

- Software Engineering | 02-24-00301
- Systems Analysis and Design | 02-24-00302
- Algorithm Design | 02-24-00303
- Distributed Processing | 02-24-00304
- Mobile Programming | 02-24-00305
- Web Programming | 02-24-00306
- Operating Systems | 02-24-00307
- Computer Networks | 02-24-00308

Required Faculty Elective Credits For Graduation: 12 Credits

=====

PROGRAM ELECTIVES

- Convex Optimization | 02-24-01409
- Non-Linear and Combinatorial Optimization | 02-24-01410
- Multivariate Statistical Analysis | 02-24-01411

- Bayesian Statistics | 02-24-01412
- Data Compression Techniques | 02-24-01413
- Concurrent Algorithms and Data Structures | 02-24-01414
- Distributed Database Systems | 02-24-01415
- Advanced Database Systems | 02-24-01416

Required Program Elective Credits For Graduation: 12 Credits

=====

UNIVERSITY REQUIREMENTS

Mandatory:

- Critical Thinking
- Innovation & Entrepreneurship

Graduation Requirements:

- Human Rights and Anti-Corruption
- Community Service and Environmental Development
- Military Education

Note:

Graduation requirements are mandatory for graduation but their credit hours are not counted in the graduation GPA.